

Assessment 1	Frontend design and usability
Due date	11.59 pm, Sunday 09 August 2026
Weighting	20%
Word count/length	900 words equivalent
Generative AI use	Full AI
Individual	This is an individual assessment.
Feedback	Written feedback will be provided after submission to support improvement for later assessments.

Purpose

This assessment is the first stage of a larger project that will continue across the subject. The project is a Wordle-style web application builder designed for Speech Pathology students and teachers, not for clients. The builder will generate HTML pages that run in a normal web browser and will allow teachers to create phoneme-based classroom activities.

Assessment 1 focuses on frontend design and usability. The purpose of this stage is to establish a clear, responsive, and user-centred interface for building and previewing phoneme-based activities. Teachers need a tool that is simple to use, flexible, and easy to understand. This assessment prepares you to apply React, usability principles, and responsive design practices to a specialist classroom tool.

MUST BE CREATED FROM: `npx create-next-app` .

Task details

Design and implement a frontend builder interface that allows teachers to create two separate phoneme-based classroom activities: a Wordle game and a Word Search activity. In Assessment 1, the focus is on building and previewing these activities as frontend outputs. The system should be designed so that teachers can configure the activity, view the result, and generate a downloadable HTML file for use in a normal web browser.

The Wordle activity should use phoneme-based words rather than standard spelling. The Word Search activity should also be built around phoneme-based content so that the teacher can create a supporting literacy or phoneme-recognition task. At this stage, the application does not need to support a database or dynamic word-list management. Those features will be introduced in later assessments.

The output of the builder should be a single HTML file that can be downloaded and run in a normal web browser. The page should display the playable Wordle or Word Search activity clearly and should support phoneme-based content, difficulty settings, and a basic workflow for generating the activity.

The application should be designed for teachers who want to prepare activities for Speech Pathology students. It should allow the teacher to choose between the Wordle and Word Search options, preview the activity, and then generate the final HTML version.

Instructions

You are required to:

1. **Design and implement a frontend builder interface** that includes:
 - component-based structure
 - responsive layout
 - clear navigation and workflow

- an interface for creating and previewing Wordle-style outputs.
- 2. **Support phoneme-based gameplay** by designing the interface around phoneme symbols rather than standard word entry.
 - The Wordle activity should use a single phoneme-based word in Assessment 1.
 - The word list and database-driven generation will be introduced in Assessment 2.
 - The Word Search activity should include a small word list of approximately five phoneme-based words and generate a word search from that list (can be fixed at this stage)
- 3. **Provide phoneme-based hints and feedback**, including:
 - mouse-over hints that show the phonetic-to-English letter equivalence
 - a clear display of the phoneme word
 - a clear display of the English equivalence when the answer is correct
 - hover labels such as /θ/ with the button label **TH (as in thin)**.
- 4. **Generate HTML output** that:
 - can run in a normal web browser
 - downloads as a single playable **.html** file
 - reflects the selected game settings
 - can be used as a Wordle-style classroom activity.
- 5. **Produce a verbal justification in the video** covering:
 - design decisions
 - component structure and scalability
 - usability considerations
 - accessibility considerations
 - trade-offs made in the frontend design
 - how the interface supports Speech Pathology students and teachers.
- 6. **Technical expectations:**
 - use React best practices
 - apply modular and reusable components
 - ensure code readability and maintainability
 - demonstrate a professional frontend suitable for later expansion.

Required pages

The application should include:

- **Home** — the main landing page, with a brief introduction to the project and links to the other pages.
- **About** — explains what the project is, confirms that Assessment 1 is frontend only, and briefly describes the Wordle and Word Search tools. This page should also include your name, student number, and a short video explaining how to use the website.
- **Wordle** — creates the Wordle game with phonemes.
- **Word Search** — creates a word search with phonemes.
- **Settings** — provides interface controls such as light/dark mode themes (stored in cookies) and optional layout preferences.

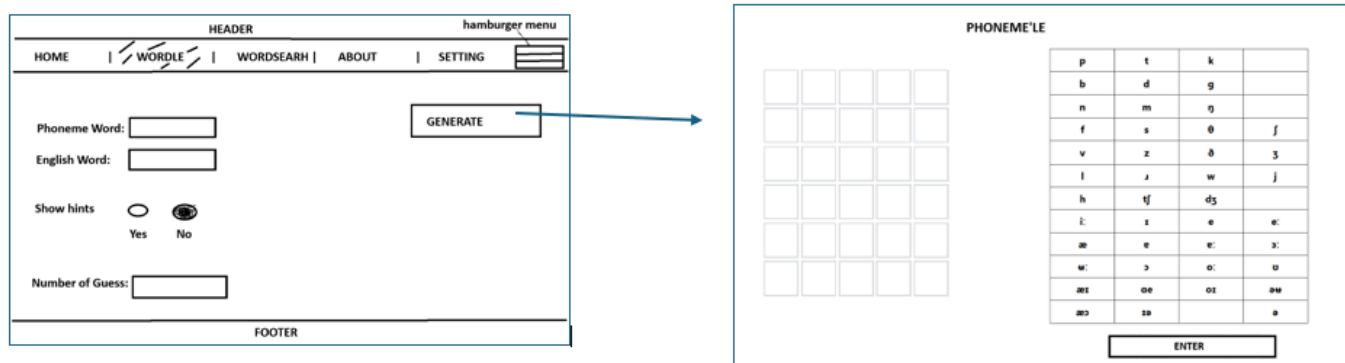


Figure of GUI. Your React App (left) generates a standalone html page (right)

Interface requirements

The interface should include:

- a **navigation bar** or **tab bar**
- a **header** with the assessment title
- a **footer** with your name and student number
- a **hamburger menu** or **kebab menu** for compact navigation, with options such as About and Settings

- a **Generate** button that downloads the playable Wordle or Word Search activity as a single `.html` file

Project continuity

This assessment forms the foundation for the broader project. Assessment 1 is intentionally focused on frontend design, usability, and user experience. Later assessments will introduce the word list, database, and more advanced generation options so that the builder can rotate through multiple phoneme words and produce richer outputs.

Resources and readings relevant to the assessment

The following resources are especially relevant to this assessment:

- React documentation
- Accessibility and usability guidelines
- Lecture and tutorial materials on frontend design, responsive layouts, and component architecture
- Course materials related to phonetics, phoneme representation, or Speech Pathology teaching tools

Assessment criteria

The rubric for this assessment will focus on frontend quality, usability, accessibility, component structure, visual design, and the clarity of the verbal justification. The detailed rubric will be developed after the assessment brief is finalised.

Submission details

Submit:

- a zip file containing your project code and GitHub repository link
- video with showing your student ID, face, and voice narration.

Remove `node_modules` before uploading if required by your submission process.

AI acknowledgement

If AI use is acceptable in this assessment, you must complete and submit an AI acknowledgement with your work. Failure to submit an acknowledgement when required may constitute a breach of academic integrity. You will find the AI acknowledgement on the Assessments page.

References (show in video if use)

- Minimum five academic or industry sources
- APA 7th ed. style

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